

2501-PYL1003 (Mechanics & Special Theory of Relativity)

Practice Mid-Semester Examination — Paper 5 (Unofficial, Instructor-Style)

Full Marks: 60 Time: 2 hours Set: Practice-5

Instructions

An MCQ can also be an MSQ (select all correct options). Justify every choice — no marks for a bare letter. Show all intermediate steps and reasoning.

State any approximation or rounding explicitly, with justification.

Numerical answers should be rounded to two decimal places unless stated otherwise.

1. In a proposed modification of Newtonian gravity, the force law is written as

$$F(r) = -\frac{GMm}{r^2} \left(1 + \frac{\alpha}{r^\gamma} \right)$$

where α is a new constant and γ is a dimensionless exponent to be determined by requiring α/r^γ to be dimensionless. (i) Find the dimension of α in terms of M, L, T for general γ (i.e., express $[\alpha]$ as a function of γ). (ii) If, on physical grounds, α is required to have dimension of L^2 (length-squared), find γ . **[4+2 = 6]**

2 (MSQ). The Reynolds number $Re = \rho v L / \eta$ and the Mach number $Ma = v / c_s$ are both dimensionless groups used in fluid mechanics (ρ : density, v : speed, L : length, η : dynamic viscosity $ML^{-1}T^{-1}$, c_s : sound speed). A new dimensionless group is proposed: $\Psi = Re^p Ma^q \left(\frac{L}{L} \right)$ — i.e., simply products of powers of already-dimensionless groups. Which of the following statement(s) is/are correct? a) Ψ is dimensionless for **any** real values of p, q , since Re and Ma are each individually dimensionless b) This reasoning would **fail** if Re or Ma were not actually dimensionless as claimed — i.e., the argument is only as good as the premise c) One can verify Re is dimensionless by direct substitution of M, L, T dimensions for each symbol d) Dimensional analysis alone can determine the correct values of p and q for a specific physical phenomenon **[5]**

3. A particle moves along a line with acceleration $a(v) = -\kappa\sqrt{v}$ (for $v > 0, \kappa > 0$ constant), starting at $v_0 > 0$ at $t = 0, x = 0$. (i) Solve for $v(t)$ and find the time t_{stop} at which $v = 0$. (ii) Find $x(t)$ for $0 \leq t \leq t_{\text{stop}}$, and hence the total distance travelled. (iii) Contrast this with the $a = -\kappa v^3$ case from Paper 4, Q3, where the particle never stops in finite time but travels finite distance. Here, does the particle stop in finite time? Briefly explain, in one or two sentences, the general connection between the power of v in

$a(v) = -\kappa v^n$ and whether t_{stop} is finite (no need for a full general proof — just state the qualitative trend based on comparing $n < 1$ here vs. $n = 3$ there). **[3+3+3 = 9]**

4. A block of mass 1.5 kg is projected up a rough incline (angle 30° with horizontal, $\mu_k = 0.25$) with initial speed $v_0 = 10$ m/sec, measured along the incline. ($g = 10$ m/sec².) (i) Find the deceleration while moving up, and the distance travelled up the incline before momentarily stopping. (ii) Determine whether the block slides back down after stopping (compare $\tan(30^\circ)$ to μ_s ; you may assume $\mu_s \approx \mu_k = 0.25$ for this check), and if it does, find its speed upon returning to the starting point. **[3+4 = 7]**

5 (MSQ). A rigid body's angular velocity vector is $\vec{\omega} = 3\hat{i} - 4\hat{j} + 12\hat{k}$ rad/sec (Cartesian). A point on the body is located, at some instant, at position $\vec{r} = 2\hat{i} + \hat{j} - \hat{k}$ (metres) relative to a point on the rotation axis. Recall $\vec{v} = \vec{\omega} \times \vec{r}$. Which of the following is/are correct about the point's velocity (two decimal places)? a) $\vec{v} = -3\hat{i} + 5\hat{j} + 11\hat{k}$ (or similar — verify by direct cross-product computation) b) $|\vec{v}|$ can be computed directly from $|\vec{\omega}||\vec{r}| \sin \Theta$ where Θ is the angle between $\vec{\omega}$ and $\vec{r}$ c) $\vec{v} \cdot \vec{\omega} = 0$ always, regardless of the specific vectors, as a general property of the cross product d) $\vec{v} \cdot \vec{r} = 0$ always, as a general property of the cross product **[6]**

6. Consider the transformation between Cartesian (x, y, z) and **parabolic cylindrical** coordinates (u, v, z) defined by

$$x = \frac{1}{2}(u^2 - v^2), \quad y = uv, \quad z = z$$

(i) Find $\hat{u}$ and $\hat{v}$ (unit vectors) in terms of $\hat{i}, \hat{j}$ and u, v , using $\hat{q} = (\partial\vec{r}/\partial q)/|\partial\vec{r}/\partial q|$ for $q = u, v$. (ii) Show that $\hat{u} \cdot \hat{v} = 0$ (i.e., this is an orthogonal coordinate system) by direct computation. (iii) Find the scale factor $h_u = |\partial\vec{r}/\partial u|$ explicitly in terms of u, v . **[4+3+2 = 9]**

7. A spacecraft is in a circular parking orbit of radius $r_1 = 7000$ km about the Earth ($GM_\oplus = 3.986 \times 10^5$ km³/sec²). It performs a **Hohmann transfer** to a higher circular orbit of radius $r_2 = 42000$ km (geostationary-like altitude), by firing its engine briefly at r_1 to enter an elliptical transfer orbit with perigee r_1 and apogee r_2 , then firing again at r_2 to circularize. (i) Find the speed on the circular orbit at r_1 , and the speed at perigee of the transfer ellipse (semi-major axis $a = (r_1 + r_2)/2$); hence find the first Δv required. (ii) Find the speed at apogee of the transfer ellipse and the speed on the final circular orbit at r_2 ; hence find the second Δv . **[4+4 = 8]**

8. A particle of mass m orbits under a central force derivable from potential $V(r) = -\frac{k}{r}e^{-r/\lambda}$ (a Yukawa-type screened potential, $k, \lambda > 0$), with angular momentum L . (i) Write the effective potential $V_{\text{eff}}(r)$. (ii) Write (do not necessarily solve in closed form) the equation $V'_{\text{eff}}(r_0) = 0$ that determines a circular-orbit radius r_0 , and show that in the limit $\lambda \rightarrow \infty$ (i.e., $r_0 \ll \lambda$) it reduces to the familiar inverse-square-orbit result $r_0 = L^2/(mk)$. **[3+5 = 8]**

9. A weather balloon at latitude $\lambda = 20^\circ$ S is caught in a purely vertical updraft, rising at constant speed $w = 8$ m/sec (relative to the ground) for a duration $\tau = 200$ sec, with no horizontal ground-relative velocity component *initially imposed by the wind itself* — but of course it will develop one due to Coriolis. Using $\Omega = 7.29 \times 10^{-5}$ rad/sec: (i) Identify which horizontal component (east-west or north-south) of the Coriolis acceleration is generated purely by a **vertical** velocity, and derive its magnitude and sign at

this southern-hemisphere latitude (careful with the sign of $\sin \lambda$ for λ south). (ii) Estimate the horizontal (East or West — state which, with reasoning) displacement accumulated over the 200 sec, using the zeroth-order (purely vertical) motion to evaluate the correction. **[5+4 = 9]**

10. A space station is constructed as a large hoop of radius $R = 200$ m, rotating about its central axis at angular velocity Ω to simulate gravity via centrifugal effect, such that a person standing on the inner rim experiences effective "weight" mg_{sim} with $g_{\text{sim}} = 9.8 \text{ m/sec}^2$. (i) Find the required Ω . (ii) An astronaut on the rim drops an object radially "downward" (i.e., releases it from rest **relative to the rotating station**, at the rim). Working in the rotating frame, write the initial (at the instant of release) centrifugal and Coriolis accelerations on the object, and determine the initial direction of the object's deflection relative to a purely radial fall (i.e., does it appear to curve toward the direction of rotation, or against it, as seen by an observer standing on the rim)? **[3+6 = 9]**

11. A raindrop falls through a stationary, uniform mist and grows by accretion such that its mass increases at a rate proportional to its **surface area**: $\frac{dm}{dt} = c m^{2/3}$ for some constant $c > 0$ (since surface area $\propto$ (radius) $^2 \propto m^{2/3}$ for constant density). The mist is at rest, so accreted mass contributes no extra momentum (i.e., there is no thrust term — treat it like the Paper 2 Q11 "leaky bucket" logic, but for **mass gain** with zero relative velocity of the gained mass). (i) Using $\frac{d}{dt}(mv) = mg$ (valid here since the added mass carries zero momentum of its own, so there is no separate thrust term — justify this briefly), and given $m(t) = (m_0^{1/3} + \frac{c}{3}t)^3$ (you may take this as given, resulting from integrating the growth law), derive the equation of motion for $v(t)$. (ii) Assuming the drop starts from rest ($v(0) = 0$) and defining $\tau(t) = m_0^{1/3} + \frac{c}{3}t$, show that the equation of motion can be written as $\frac{d}{dt}(\tau^3 v) = g\tau^3$, and hence find $v(t)$ in terms of $g, c, \tau(t), m_0$. **[3+6 = 9]**

12. A cart of mass M (excluding sand) moves on a frictionless horizontal track. Sand is poured into the cart from a stationary hopper directly above, at a constant rate σ (kg/sec), landing with zero horizontal velocity (relative to the ground) each time. The cart has initial mass M (empty) and initial speed v_0 at $t = 0$ (given an initial push, then left alone — no further horizontal force applied by any motor). (i) Using conservation of horizontal momentum for the cart+accumulated-sand system (since the incoming sand carries zero horizontal momentum and there's no external horizontal force), find $v(t)$ in terms of M, σ, v_0, t . (ii) Find the kinetic energy of the system at time t , and show explicitly that it is **less** than $\frac{1}{2}Mv_0^2$ (the initial KE) for $t > 0$ — identify physically where the "missing" energy has gone. **[4+5 = 9]**

13. A uniform rod of length L and mass M is hinged at one end to a wall (frictionless hinge) and held horizontal by a light cable attached to the rod's far end, making angle $\theta = 40^\circ$ with the rod (the cable runs from the rod's free end up to a point on the wall above the hinge). A mass m hangs from a point at distance d from the hinge along the rod ($0 < d < L$). (i) Taking torques about the hinge, derive an expression for the tension T in the cable in terms of M, m, g, L, d, θ . (ii) Given $M = 8 \text{ kg}, m = 5 \text{ kg}, L = 3 \text{ m}, d = 2 \text{ m}, \theta = 40^\circ$ ($\sin 40^\circ \approx 0.643$), find T and also find the horizontal and vertical components of the force exerted by the hinge on the rod ($g = 10 \text{ m/sec}^2$). **[4+5 = 9]**

14 (MSQ). A solid uniform sphere of mass m and radius a rolls without slipping down an incline of angle θ . (Moment of inertia of a solid sphere about its center: $I = \frac{2}{5}ma^2$.) Using Newton's second law for

translation and rotation (torque about the center) separately, with f the (unknown, to be solved for) friction force: a) The linear acceleration down the incline is $a_{cm} = \frac{5}{7}g \sin \theta$ b) The required friction force is $f = \frac{2}{7}mg \sin \theta$, directed up the incline c) Rolling without slipping requires $\mu_s \geq \frac{2}{7} \tan \theta$ d) If the sphere were replaced by a hollow spherical shell ($I = \frac{2}{3}ma^2$), the linear acceleration down the incline would be **greater** than for the solid sphere **[6]**

15. Two particles of masses m_1 and m_2 interact solely via a mutual central force $F(r) = -k(r - r_0)$ for r near r_0 (an attractive linear restoring force about equilibrium separation r_0 , valid for small deviations — think of them connected by an ideal spring with natural separation r_0 , but free to also revolve about their common center of mass, i.e., a genuine two-body central-force problem, not a fixed-axis problem). (i) Reduce the two-body problem to an equivalent one-body problem using the reduced mass $\mu = m_1 m_2 / (m_1 + m_2)$ and relative coordinate $\vec{r} = \vec{r}_1 - \vec{r}_2$; write the effective one-body radial equation of motion including the centrifugal term (angular momentum L of the relative motion). (ii) Find the equilibrium separation r_c (circular-orbit radius, generally $\neq r_0$ because of the centrifugal term) in terms of k, r_0, L, μ **to first order**, assuming the centrifugal contribution is a small perturbation (i.e., write $r_c = r_0 + \delta$ and find δ to leading order in L^2). **[4+5 = 9]**

End of Paper 5. Total: 60 marks.